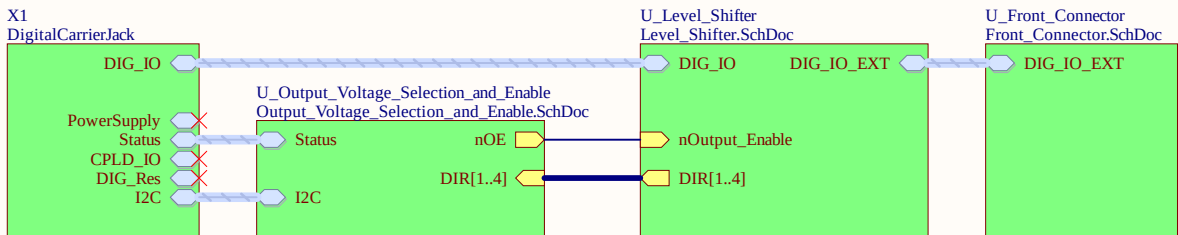
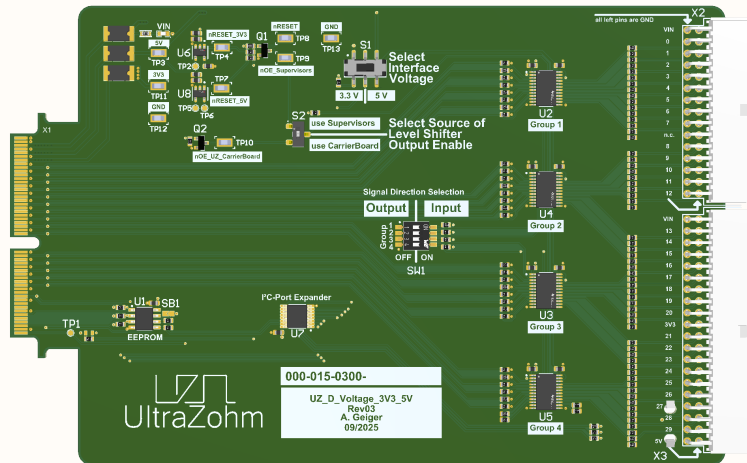
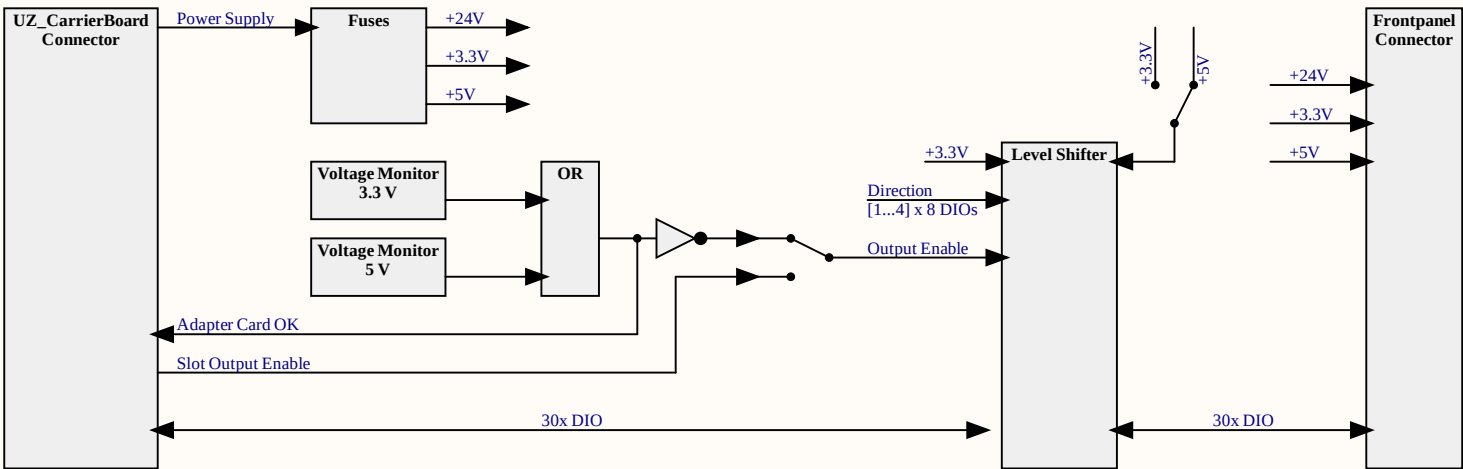


Top Sheet

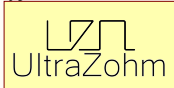


Block Diagram

- Board Features:
- configurable output voltage level (3.3V or 5V)
 - voltage monitor ICs supervise 3.3V and 5V supply rail
 - output enable of level shifters can be controlled either by the supervisors status or via UZ_CarrierBoard signal
 - I²C-Port Expander reads selected output voltage level and signal direction per group of 8 channels
 - RGB-LED at front panel visualizes selected voltage and error status



Chose size of UZ logo by choosing footprint type



LOGO1

INFO1

Project
ProjectRevision
AuthorParam
ProjectDate

Design Information

INFO2

ZC_GGG
ZC_PPP
ProjectRevision
ZC_vv

ZC Serialnumber

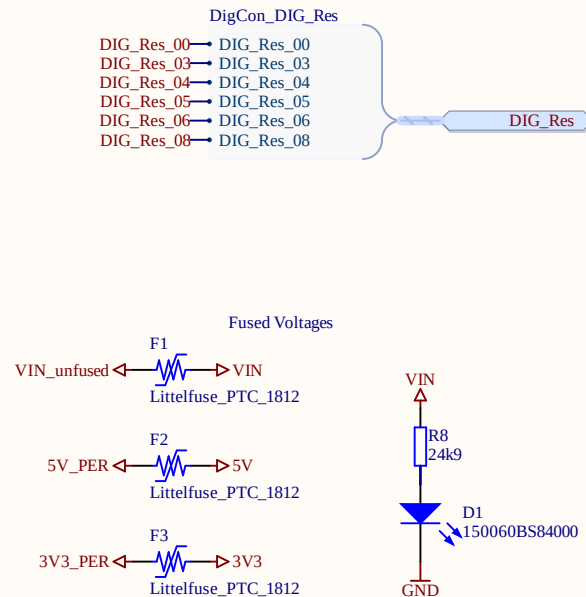
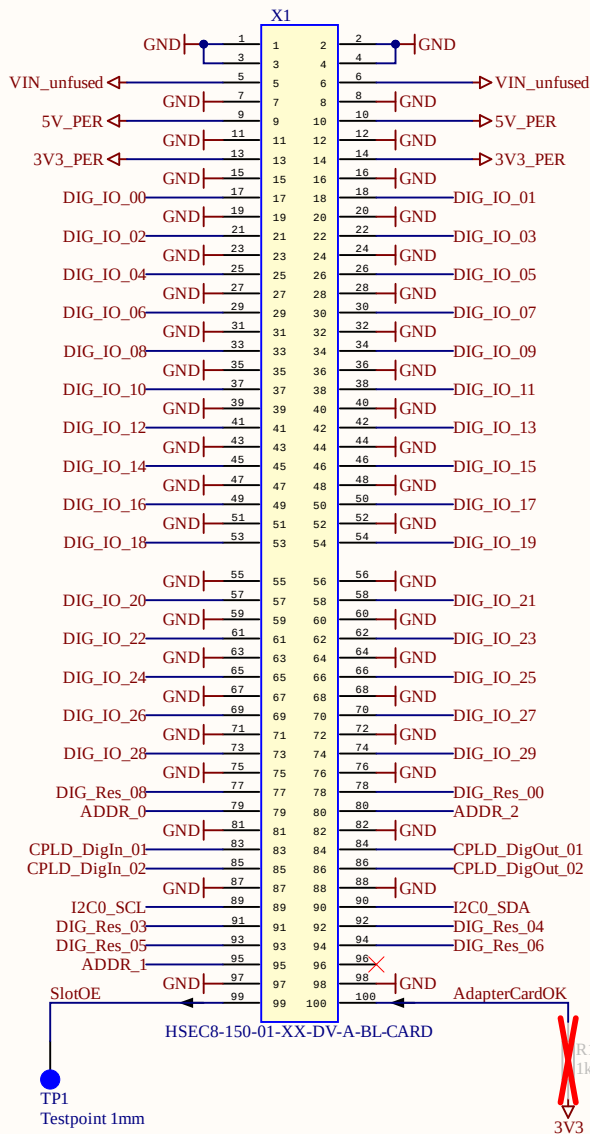
Title TopSheet.SchDoc		UltraZohm www.ultrazohm.com		
Revision: 03	Design Engineer: A. Geiger			
Project: UZ_D_Voltage_3V3_5V.PrjPCB				Date: 20.11.2025
				Sheet 1 of 5

A

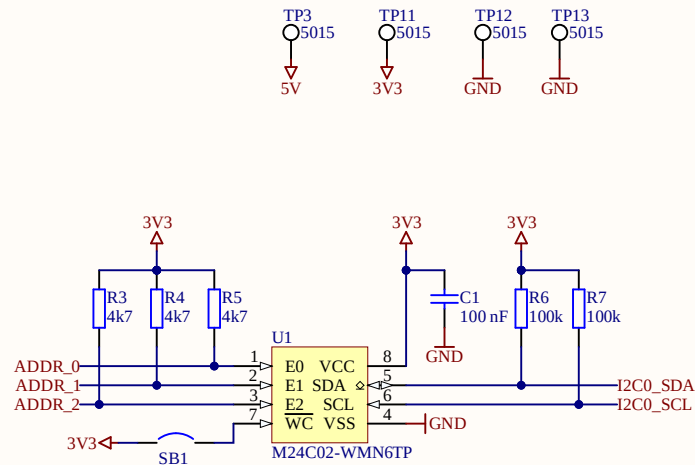
B

C

D



Debugging Testpoints



Note: place solder bump to protect EEPROM memory from write access.

PC-Address:
0b101 E2 E1 E0 = 0h5[0-7]

E2/E1/E0 is configured on UZ_CarrierBoard hardware related to the used slot D[1..5].

Title DigitalCarrierJack.SchDoc

Revision: 03

Design Engineer: A. Geiger

Project: UZ_D_Voltage_3V3_5V.PrjPCB

UltraZohm

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Date: 20.11.2025

Sheet 2 of 5

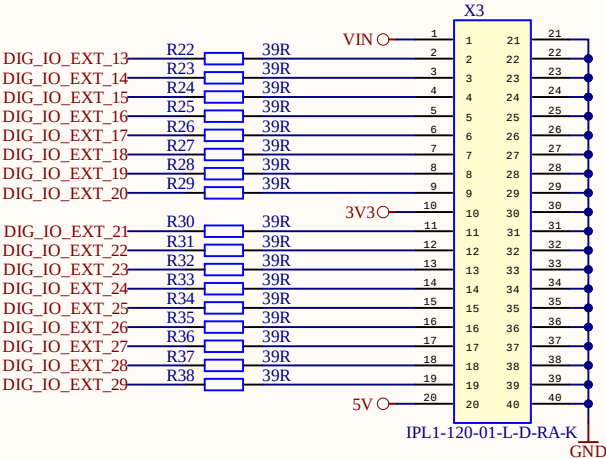
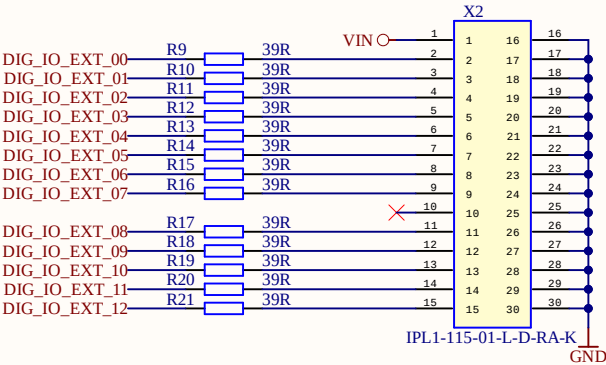
Pull-Down Resistors

Front Panel Connector

DigCon_DIG_IO_EXT

DIG_IO_EXT_00 → DIG_IO_EXT_00
DIG_IO_EXT_01 → DIG_IO_EXT_01
DIG_IO_EXT_02 → DIG_IO_EXT_02
DIG_IO_EXT_03 → DIG_IO_EXT_03
DIG_IO_EXT_04 → DIG_IO_EXT_04
DIG_IO_EXT_05 → DIG_IO_EXT_05
DIG_IO_EXT_06 → DIG_IO_EXT_06
DIG_IO_EXT_07 → DIG_IO_EXT_07
DIG_IO_EXT_08 → DIG_IO_EXT_08
DIG_IO_EXT_09 → DIG_IO_EXT_09
DIG_IO_EXT_10 → DIG_IO_EXT_10
DIG_IO_EXT_11 → DIG_IO_EXT_11
DIG_IO_EXT_12 → DIG_IO_EXT_12
DIG_IO_EXT_13 → DIG_IO_EXT_13
DIG_IO_EXT_14 → DIG_IO_EXT_14
DIG_IO_EXT_15 → DIG_IO_EXT_15
DIG_IO_EXT_16 → DIG_IO_EXT_16
DIG_IO_EXT_17 → DIG_IO_EXT_17
DIG_IO_EXT_18 → DIG_IO_EXT_18
DIG_IO_EXT_19 → DIG_IO_EXT_19
DIG_IO_EXT_20 → DIG_IO_EXT_20
DIG_IO_EXT_21 → DIG_IO_EXT_21
DIG_IO_EXT_22 → DIG_IO_EXT_22
DIG_IO_EXT_23 → DIG_IO_EXT_23
DIG_IO_EXT_24 → DIG_IO_EXT_24
DIG_IO_EXT_25 → DIG_IO_EXT_25
DIG_IO_EXT_26 → DIG_IO_EXT_26
DIG_IO_EXT_27 → DIG_IO_EXT_27
DIG_IO_EXT_28 → DIG_IO_EXT_28
DIG_IO_EXT_29 → DIG_IO_EXT_29

DIG_IO_EXT_00 R42 100k GND
DIG_IO_EXT_01 R73 100k GND
DIG_IO_EXT_02 R74 100k GND
DIG_IO_EXT_03 R85 100k GND
DIG_IO_EXT_04 R86 100k GND
DIG_IO_EXT_05 R87 100k GND
DIG_IO_EXT_06 R88 100k GND
DIG_IO_EXT_07 R89 100k GND
DIG_IO_EXT_08 R90 100k GND
DIG_IO_EXT_09 R91 100k GND
DIG_IO_EXT_10 R92 100k GND
DIG_IO_EXT_11 R93 100k GND
DIG_IO_EXT_12 R94 100k GND
DIG_IO_EXT_13 R95 100k GND
DIG_IO_EXT_14 R96 100k GND
DIG_IO_EXT_15 R97 100k GND
DIG_IO_EXT_16 R98 100k GND
DIG_IO_EXT_17 R99 100k GND
DIG_IO_EXT_18 R100 100k GND
DIG_IO_EXT_19 R101 100k GND
DIG_IO_EXT_20 R102 100k GND
DIG_IO_EXT_21 R103 100k GND
DIG_IO_EXT_22 R104 100k GND
DIG_IO_EXT_23 R105 100k GND
DIG_IO_EXT_24 R106 100k GND
DIG_IO_EXT_25 R107 100k GND
DIG_IO_EXT_26 R108 100k GND
DIG_IO_EXT_27 R109 100k GND
DIG_IO_EXT_28 R110 100k GND
DIG_IO_EXT_29 R111 100k GND



Title Front_Connector.SchDoc

Revision: 03

Design Engineer: A. Geiger

Project: UZ_D_Voltage_3V3_5V.PrjPCB

UltraZohm

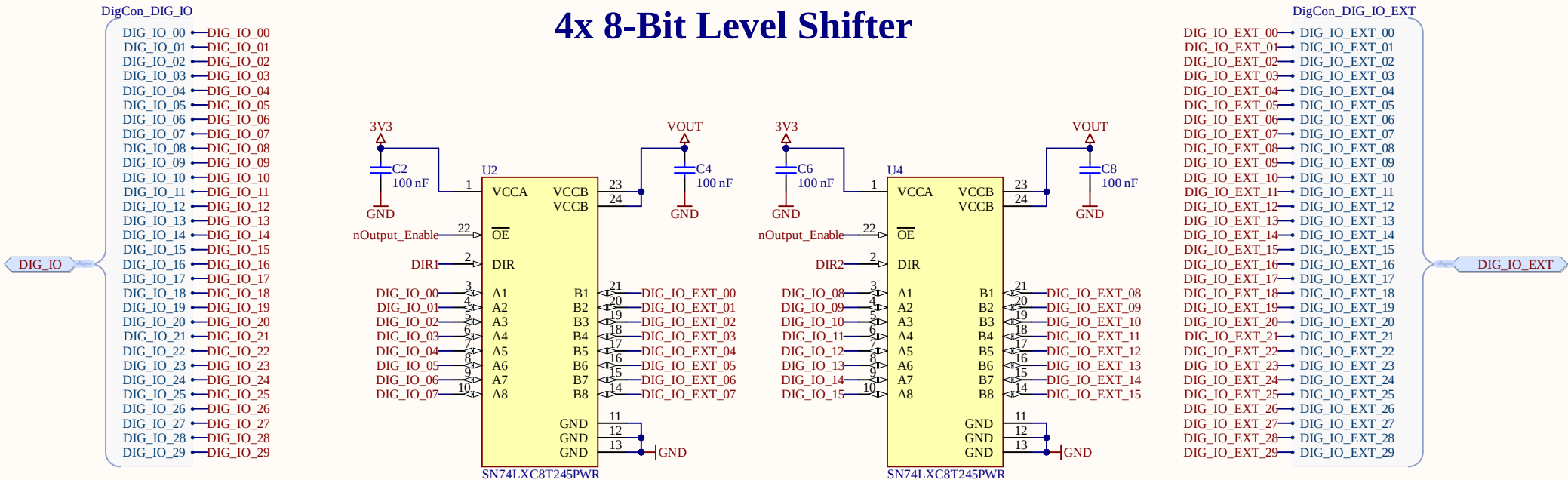
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Date: 20.11.2025

Sheet 3 of 5

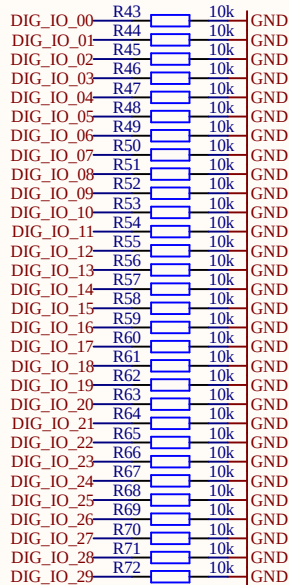


4x 8-Bit Level Shifter



nOutput_Enable

Pull-Down Resistors



Direction per Level Shifter

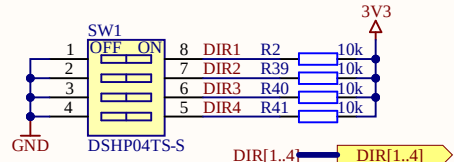


Table 8-1. Function Table ⁽¹⁾			
CONTROL INPUTS		Port Status	
OE	DIR	A PORT	B PORT
L	L	Output (Enabled)	Input (Hi-Z)
L	H	Input (Hi-Z)	Output (Enabled)
H	X	Input (Hi-Z)	Output (Hi-Z)

(1) Input circuits of the data I/Os are always active and should be kept at a valid logic level.

DIP-Switch	DIR	Signal Direction
OFF	HIGH	Output
ON	LOW	Input

